



Please show me how the Binomial Theorem can be used to produce the Taylor series for the exponential function.

The Binomial Theorem provides a path to the Taylor series of e^x by expressing the exponential function as the limit of a compounded growth process: $e^x = \lim_{n \rightarrow \infty} (1 + \frac{x}{n})^n$.

1. Apply the Binomial Theorem

We begin by expanding the term $(1 + \frac{x}{n})^n$ using the general formula for the Binomial Theorem:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

Setting $a = 1$ and $b = \frac{x}{n}$, we get:

$$\left(1 + \frac{x}{n}\right)^n = \sum_{k=0}^n \binom{n}{k} \left(\frac{x}{n}\right)^k$$

2. Expand the Binomial Coefficient

To see the behavior as n grows, we expand the binomial coefficient $\binom{n}{k} = \frac{n!}{k!(n-k)!}$:

$$\binom{n}{k} = \frac{n(n-1)(n-2)\dots(n-k+1)}{k!}$$

Ask anything





$$\left(1 + \frac{x}{n}\right)^n = \sum_{k=0}^n \frac{x^k}{k!} \left[\frac{n}{n} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n} \dots \frac{n-k+1}{n} \right]$$

Which simplifies to:

$$\left(1 + \frac{x}{n}\right)^n = \sum_{k=0}^n \frac{x^k}{k!} \left[1 \cdot \left(1 - \frac{1}{n}\right) \cdot \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{k-1}{n}\right) \right]$$

4. Take the Limit as $n \rightarrow \infty$

As n approaches infinity, every term of the form $\frac{j}{n}$ (where j is a constant) vanishes toward zero. Therefore:

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) = 1, \quad \lim_{n \rightarrow \infty} \left(1 - \frac{2}{n}\right) = 1, \quad \dots$$

The entire product inside the brackets becomes $1 \cdot 1 \cdot 1 \dots = 1$. The upper limit of the summation n also goes to infinity. [↗](#)

✓ Answer

The process results in the standard **Taylor series for the exponential function**:

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Would you like to see how this same logic applies to deriving the series for $(1+x)^n$ when n is not an integer?





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